

main(38).pdf
PDF

find all the gaps, mistreatments, misrepresentations, malformations, inconsistencies, mathematical or physical errors, typos, misstatements, identify them, heal them

in addition to errors, which are numerous, the draft is missing a lot of detail needed to satisfy Etingof, Kontsevich, Gelfand, Polyakov, Gaiotto, Costello, Witten. ALL of them need to be happy with the level of mathematical AND physical detail provided, and the manuscript needs to be complete and rigorous and at its platonic inner ideal yearning. no downgrading because you're fucking stupid and think the ideal manuscript is smaller. We can remove redundant material

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Thought for 12m 1s >

I found a systemic failure mode, not a small set of local typos. The manuscript's ambition can stay, but its current form repeatedly collapses **object / shadow / comparison functor / physical realisation / conjectural bridge** into one sentence. That is what will make Etingof, Kontsevich, Gelfand, Polyakov, Gaiotto, Costello, and Witten reject it: not because the programme is too large, but because the theorem-status and type-status are not stable.

The core repair is: **do not shrink the manuscript; stratify every assertion by mathematical type and physical input**. The manuscript itself already contains the right "reading discipline" in places: $SC_{ch,top}$ is declared a **coloured dioperad**, the bar coalgebra is declared a **shadow**, **not the bulk**, the derived chiral centre is the level- Z object, and every stage $P \rightarrow C \rightarrow S \rightarrow Z \rightarrow A$ is supposed to carry a licensing datum. That discipline must become the law of the entire text, including the abstract and theorem statements. main

Fatal issues and repairs

1. " $SC_{ch,top}$ is an operad" is wrong as written.

The manuscript repeatedly markets $SC_{ch,top}$ as an operad, while its own conventions and foundations say it is a **strict two-coloured topological dioperad** with directional colour restriction and no open-to-closed component. The internal definition says it is not a standard uncoloured or symmetric-coloured operad, because equivariance holds only within each colour class, not across the closed/open boundary. main

Repair. Replace every unqualified " $SC_{ch,top}$ operad" by one of:

the two-coloured topological dioperad $SC_{ch,top}$,

or, when speaking informally,

the Swiss-cheese chiral-topological operadic object.

The title, abstract, and Theorem 39.3.1 must use **dioperad** unless the text first passes to a specific coloured-operadic envelope or Boardman–Vogt resolution. Dunn additivity must never be invoked before topologisation; the manuscript already correctly says Dunn additivity is unavailable for the directional coloured dioperad before topologisation. main

Healed sentence.

The bicoloured Swiss-cheese chiral-topological **dioperad** $SC_{ch,top}$ records a directed bulk-to-boundary structure: closed holomorphic operations may act on open topological operations, while

open inputs never compose into closed outputs.

2. The master theorem is overtyped: a chiral algebra alone does not canonically produce a 3d QFT.

The abstract claims a canonical functor

$$\Phi_{\text{hol}} : \text{ChirAlg}_X^{\omega, BL} \rightarrow \text{HT-QFT}_{X \times \mathbb{R}}.$$

But a chiral algebra with conformal vector and "boundary linearisation" is not enough data to construct a physical 3d holomorphic-topological QFT. The manuscript later admits the needed data: BV datum, product HT gauge fixing, factorised logarithmic parametrix, kernel homotopy, product-formal local-shadow recognition, and comparison hypotheses. main

Repair. Split the theorem into two functors:

$$\mathfrak{Z}_{\text{ch}} : \text{ChirAlg}_X^{\text{Kosz}} \longrightarrow \text{SC}_{\text{ch, top}}\text{-Alg}_{\text{alg}}$$

is the **algebraic derived-centre construction**, while

$$\Phi_{\text{hol}} : \text{HTReal}_X^{\omega, BL, \Xi} \longrightarrow \text{HT-QFT}_{X \times \mathbb{R}}$$

is the **physical realisation functor**, whose source includes the HT/BV realisation datum Ξ .

Healed theorem statement.

Let (A, ω, BL, Ξ) be a chirally Koszul chiral algebra on a smooth curve X , equipped with a non-critical conformal vector, a boundary linearisation, and HT realisation data Ξ consisting of a BV field complex, gauge fixing, factorised logarithmic parametrix, anomaly neutralisation, and the product-formal recognition datum of the $\text{SC}_{\text{ch, top}}$ -algebra. Then the observables of the associated HT realisation $T_{A, \Xi}$ satisfy

$$\text{Obs}_{\partial}(T_{A, \Xi}) \simeq A, \quad \text{Obs}_{\text{bulk}}(T_{A, \Xi}) \simeq C_{\text{ch}}^{\bullet}(A, A)$$

as $\text{SC}_{\text{ch, top}}$ -algebras. If, in addition, the holomorphic translations are made Q -exact by the specified antighost/topologisation datum, then $C_{\text{ch}}^{\bullet}(A, A)$ upgrades to the stated E_3^{top} -factorisation algebra. For class $\mathcal{M}$, this assertion is in the weight-completed/pro-Banach ambient unless finite propagation on the original direct-sum complex is supplied.

That keeps the ambition and makes the theorem true-shaped.

3. The derived centre is misdefined in the abstract.

The displayed formula

$$Z_{\text{ch}}^{\text{der}}(A) = R\text{Hom}_{A \otimes A^{\text{op}}}(A, A)$$

is ordinary associative Hochschild notation. For a chiral algebra, the object must be defined in the **chiral/Ran/factorisation bimodule category**; only after a comparison functor may it resemble ordinary Hochschild cochains.

Repair. Define

$$Z_{\text{ch}}^{\text{der}}(A) := \text{REnd}_{A\text{-bimod}}^{\text{ch}}(A) \simeq C_{\text{ch}}^{\bullet}(A, A),$$

where $A\text{-bimod}$ means the appropriate chiral/Ran A -bimodule category. Then add a comparison lemma explaining when this agrees with an ordinary $R\text{Hom}_{A \otimes A^{\text{op}}}(A, A)$ -model.

Healed sentence.

The chiral derived centre is the derived endomorphism object of A as a chiral A -bimodule on the Ran space:

$$Z_{\text{ch}}^{\text{der}}(A) := \text{REnd}_{A\text{-}A}^{\text{ch}}(A).$$

A presentation by an ordinary Hochschild $RHom$ is a model-dependent comparison, not the definition.

4. The manuscript confuses E_2 , $SC_{\text{ch},\text{top}}$, and E_3^{top} .

The derived centre of an E_1 -type algebra is naturally E_2 -like by Deligne/Tamarkin. The $SC_{\text{ch},\text{top}}$ datum lives on the pair $(Z_{\text{ch}}^{\text{der}}(A), A)$. The E_3^{top} structure is a further physical/topological promotion after Q -exact holomorphic translations. The manuscript itself says the mixed chiral-topological structure is on the **derived-centre pair**, not on A alone. main

Repair. Every occurrence of " A is E_3 -topological" should become either:

$$Z_{\text{ch}}^{\text{der}}(A) \text{ is } E_3^{\text{top}}$$

under topologisation hypotheses, or

$$(Z_{\text{ch}}^{\text{der}}(A), A) \text{ is an } SC_{\text{ch},\text{top}}\text{-algebra}$$

without them.

5. "Generic Virasoro/W-algebras are C_2 -cofinite" is false.

The manuscript uses "non-logarithmic C_2 -cofinite standard landscape" while including generic Virasoro and generic W_3 . Generic Virasoro representation theory is instead treated through C_1 -cofinite finite-length categories; generic Virasoro categories can be semisimple and rigid in that setting, but that is not the same as C_2 -cofiniteness. 🔗 [MS Researchers](#) Also, C_2 -cofiniteness implies finite-dimensional Zhu algebra, but finite-dimensional Zhu algebra is not equivalent to C_2 -cofiniteness in general; the literature states the implication C_2 -cofinite $\Rightarrow$ finite-dimensional Zhu, not the converse. ✖ [arXiv +1](#)

Repair. Replace the landscape label with:

the non-logarithmic tempered/non-critical standard landscape

and reserve C_2 -cofinite for minimal/admissible/rational/lisse/holomorphic loci where it is actually proved.

Healed sentence.

The standard landscape splits into C_2 -cofinite rational/lisse loci, generic non-critical tempered loci, and logarithmic loci. Generic Virasoro and generic principal W_N belong to the tempered non-critical locus, not to the C_2 -cofinite locus.

6. The A_∞ /Fulton–MacPherson geometry is notationally inconsistent.

The manuscript alternates among $FM_n(\mathbb{C})$, reduced $FM_n(\mathbb{C})/\text{Aff}(\mathbb{C})$, and translation-normalised configuration spaces without a stable notation. It says $FM_n(\mathbb{C})/\text{Aff}(\mathbb{C})$ has real dimension $2n - 4$, but later uses quotient coordinates $w_i = z_i - z_4$ and says $FM_4(\mathbb{C})$ has real dimension 6, which is only translation-normalised, not full affine-reduced. main

Repair. Introduce three distinct symbols:

$FM_X(n)$ unreduced compactification on the curve,

$FM_{\text{tr}}(n) := FM_X(n)/\mathbb{C}$ translation-reduced,

$FM_{\text{red}}(n) := FM_X(n)/(\mathbb{C} \rtimes \mathbb{C}^\times)$ affine-reduced.

Then never write $FM_n(\mathbb{C})$ without saying which one is used.

Healed convention.

All Stokes computations for planar A_∞ operations use $FM_{\text{red}}(n)$ when the global translation and dilation/rotation have been quotiented. Computations using coordinates $w_i = z_i - z_n$ are only translation-reduced and live in $FM_{\text{tr}}(n)$.

7. Closed chiral collisions and planar A_∞ collisions are being mixed.

In the $n = 3$ and $n = 4$ A_∞ verifications, the manuscript sometimes kills nonconsecutive divisors by "ordering constraint." That is valid only after projecting to the **planar/open E_1 sector**. It is not valid for the full closed chiral $FM_n(\mathbb{C})$ colour, where all labelled collision strata exist. main

Repair. Each Stokes proof must declare the sector:

- closed symmetric chiral sector: all collision strata;
- open planar E_1 sector: only interval/consecutive strata;
- mixed sector: product with explicit projection killing selected strata.

Healed sentence.

The divisor $D_{\{1,3\}}$ does not vanish in the closed chiral $FM_3(\mathbb{C})$ calculus. It vanishes only after applying the planar open-colour projection to the ordered E_1 sector. The proof below is therefore a proof in the planar open projection, not in the full closed chiral colour.

8. The physical gravity claim is too strong unless explicitly chiralised.

The manuscript's Virasoro $\Rightarrow$ 3d gravity claim must be phrased as a **chiral/holomorphic sector** of AdS_3 gravity unless the second Virasoro copy, real structure, modular invariance, and saddle data are supplied. Brown–Henneaux gives $c = 3\ell/(2G)$ for the asymptotic Virasoro central charge in AdS_3 gravity.

INSPIRE The modern Virasoro TQFT programme is explicitly similar to, but not equivalent to, $SL(2, \mathbb{R})$ Chern–Simons theory; it is a Teichmüller/Virasoro TQFT reformulation of negative-cosmological-constant 3d gravity. X arXiv

Repair.

$A = \text{Vir}_c$ gives the chiral holomorphic boundary sector of the HT avatar of AdS_3 gravity. Full nonchiral pure gravity requires $\text{Vir}_c \boxtimes \text{Vir}_c$, a real form, Brown–Henneaux boundary conditions, modular invariance, and a specified saddle or topology-sum prescription.

This preserves the gravity ambition and removes the false implication that one chiral algebra alone is the full gravitational path integral.

9. The Costello–Francis–Gwilliam 2026 comparison contradicts itself.

Appendix B correctly says CFG 2026 concerns ordinary topological Chern–Simons factorisation algebras and knot invariants, not 6d holomorphic Chern–Simons, K3 compactification, or a BKM boundary algebra.

main That matches the actual CFG 2026 abstract: it constructs a filtered E_3 -algebra by BV quantisation of ordinary Chern–Simons theory and recovers Reshetikhin–Turaev link invariants. X arXiv
But Chapter 34 says every row is an equivalence and the "Difference" column is empty, which is mathematically false and contradicts Appendix B. main

Repair. Delete the "every row is an equivalence / difference none" language. Replace it with Appendix B's scoped comparison: BV-template analogy, hCS-lane comparison, reduced E_3 -recognition only after local constancy/topological reduction, and no equality of factorisation algebras before the comparison functor.

main

10. The bibliography has current-reference defects.

The manuscript cites "Virasoro TQFT and Teichmüller theory, arXiv:2311.15063" in the bibliography

snippet. The relevant Collier–Eberhardt–Zhang works are “Solving 3d Gravity with Virasoro TQFT,” arXiv:2304.13650, and “3d gravity from Virasoro TQFT: Holography, wormholes and knots,” arXiv:2401.13900.  arXiv +1

Repair. Replace the bibliographic entry with the actual titles and arXiv numbers. Do not cite talks or informal titles as theorem sources unless the claim is explicitly labelled “lecture-note input” or “physical motivation.”

11. The manuscript has unresolved references and leaked labels.

A mechanical scan of the PDF text finds hundreds of unresolved or raw-source artefacts: “Theorem)”, “Vol I Remark)”, “Conjecture .”, and raw labels such as `thm: , chap: , conj: .` The file excerpts show these artefacts directly in mathematical prose, for example raw `thm:koszul-reflection , chap:climax-platonic , and conj:tempered-stratum-contains-wp .` `main`

Repair. No theorem should be read until the source compiles with zero unresolved references. Run a hard reference audit:

```
latexmk -pdf -halt-on-error -interaction=nonstopmode main.tex
grep -R "Theorem )\|Conjecture \.\|Vol I Remark )\|thm:\|chap:\|conj:\|eq:" -n .
```

Every raw label must become either a resolved reference, a bibliographic citation, or an explicit “unresolved internal reference” note.

Replacement for the abstract’s first theorem block

Use this as the new abstract-level master statement.

Let X be a smooth complex curve. Let A be a chirally Koszul chiral algebra on X , equipped with a non-critical conformal vector ω . Its chiral derived centre is

$$Z_{\text{ch}}^{\text{der}}(A) := \text{REnd}_{A-A}^{\text{ch}}(A) \simeq C_{\text{ch}}^{\bullet}(A, A).$$

The pair

$$(Z_{\text{ch}}^{\text{der}}(A), A)$$

carries a natural algebra structure over the directed two-coloured chiral-topological dioperad $SC_{\text{ch}, \text{top}}$. The closed colour is controlled by holomorphic Fulton–MacPherson collision strata on X ; the open colour is controlled by the ordered E_1 -topological direction. This structure is algebraic and does not by itself assert the existence of a physical QFT.

If A is additionally equipped with HT realisation data Ξ —a BV field theory on $X \times \mathbb{R}_{\geq 0}$, a boundary condition with boundary algebra A , a factorised logarithmic parametrix, anomaly neutralisation, and the product-formal recognition datum—then the corresponding HT theory $T_{A, \Xi}$ has

$$\text{Obs}_{\partial}(T_{A, \Xi}) \simeq A, \quad \text{Obs}_{\text{bulk}}(T_{A, \Xi}) \simeq Z_{\text{ch}}^{\text{der}}(A)$$

in the scoped comparison category. If the conformal vector is made Q -exact by a specified antighost/topologisation datum, the bulk obtains the advertised E_3^{top} -factorisation structure. For class M , this statement is made in the weight-completed/pro-Banach ambient unless finite propagation on the original complex is separately proved.

That is the same ambition, but typed correctly.

Global healing rule

Every theorem must start with a status header:

Status: Proved here / Imported / Conditional / Conjectural / Analogy.

Then list:

Ambient: ordinary chain / derived / coderived / pro / Banach / weight-completed.

Then list:

Inputs: algebraic, analytic, BV, anomaly, modular, physical.

The manuscript already has the right idea in its “Results by scope” and “Constitutional concordance” sections: algebraic $SC_{ch,top}$ -results are unconditional once a logarithmic $SC_{ch,top}$ -algebra is given; physical identifications require HT realisation hypotheses; global Ran-space or modular claims require additional comparison data. main

The next serious pass should rewrite the abstract, Preface, Chapter 0.4, Chapter 2, Chapter 34, Chapter 39, and Chapter 47.9 under this rule before polishing individual computations.

          Sources

this is missing a lot of detail needed to satisfy etingof, kontsevich, gelfand, polyakov, gaiotto, costello, witten. ALL of them need to be happy with the level of mathematical AND physical detail provided don't ever say "convention layer" is that how witten? costello? gaiotto? would write? what did I tell you to do?

this is missing a lot of detail needed to satisfy etingof, kontsevich, gelfand, polyakov, gaiotto, costello, witten. ALL of them need to be happy with the level of mathematical AND physical detail provided. EVERY WORD WRITTEN SHOULD BE LIKE ONE OF THE MASTERS WROTE IT and every sentence should be pure mathematics or pure

Show more 

Thought for 5m 59s >

You told me to **heal the manuscript without lowering its ambition**, and to make every sentence carry mathematical or physical content at the level demanded by Etingof, Kontsevich, Gelfand, Polyakov, Gaiotto, Costello, and Witten. The repair is not a shorter manuscript. The repair is a manuscript in which every claimed equivalence is a functor with hypotheses, every physical phrase is a BV/factorisation-algebra construction, every operadic object has the correct colour and variance, every gravity statement states which path integral or chiral sector it names, and every shadow is separated from the object it shadows.

Below is the replacement spine.

I. The first page must be rewritten as mathematics, not programme prose

The abstract currently defines

$$Z_{ch}^{der}(A) = RHom_{A \otimes A^{op}}(A, A)$$

and then sends A by a canonical functor

$$\Phi_{hol} : ChirAlg_X^{\omega, BL} \rightarrow HT-QFT_{X \times \mathbb{R}}.$$

That sentence is not typed correctly. The manuscript itself later supplies the missing datum Ξ : the physical realisation datum, the bicoloured primitive presentation, the boundary comparison, and the bulk Hochschild comparison. Chapter 41 states the correct source as an admissible boundary-linear category of chiral algebras with conformal vector **and realisation datum** Ξ , and it says the functor is physical only after that datum is supplied. main

Replace the opening theorem by:

$$\Phi_{\text{hol}} : \mathcal{HTReal}_X^{Kosz, \omega, BL} \longrightarrow \mathcal{HTQFT}_{X \times \mathbb{R}_{\geq 0}}.$$

An object of $\mathcal{HTReal}_X^{Kosz, \omega, BL}$ is a tuple

$$(A, \omega, BL, \mathcal{E}, Q, I, \omega_{BV}, \mathcal{L}_\partial, P_L, \eta_\partial, \chi_{\text{bulk}}, G_\omega, h_{\text{HT}})$$

with the following data.

$$A \in \text{ChAlg}(X)$$

is a chiral algebra in the Beilinson–Drinfeld category of right D_X -modules on the Ran space.

$$\omega \in A$$

is a conformal vector at non-critical level.

$$(\mathcal{E}, Q, I, \omega_{BV})$$

is an elliptic BV theory on $X \times \mathbb{R}_{\geq 0}$, where

$$\mathcal{E} = \Gamma(X \times \mathbb{R}_{\geq 0}, E)$$

is a cochain complex of fields, Q is the linearised BRST operator, $I \in \mathcal{O}_{\text{loc}}(\mathcal{E})$ is the interaction, and

$$(Q + \hbar \Delta_L) e^{I[L]/\hbar} = 0$$

is the renormalised quantum master equation at scale L .

$$\mathcal{L}_\partial \subset \mathcal{E}|_\partial$$

is a BV Lagrangian boundary condition, and

$$BL : A \longrightarrow \text{Obs}_\partial^q$$

is a quasi-isomorphism of boundary chiral factorisation algebras.

$$\chi_{\text{bulk}} : Z_{\text{ch}}^{\text{der}}(A) \longrightarrow \text{Obs}_{\text{bulk}}^q$$

is a quasi-isomorphism of bulk factorisation algebras in the declared ambient.

$$G_\omega$$

is an antighost for the conformal vector:

$$T_\omega = [Q, G_\omega].$$

The homotopy h_{HT} makes the holomorphic translations Q -exact:

$$\partial_z = [Q, h_z], \quad \partial_{\bar{z}} = [Q, h_{\bar{z}}]$$

in the chosen ordinary, completed, or pro-Banach chain model.

Then the theorem becomes:

$$\text{Obs}_\partial^q(\Phi_{\text{hol}}(A, \Xi)) \simeq A, \quad \text{Obs}_{\text{bulk}}^q(\Phi_{\text{hol}}(A, \Xi)) \simeq Z_{\text{ch}}^{\text{der}}(A)$$

as $SC_{\text{ch}, \text{top}}$ -algebras. If $T_\omega = [Q, G_\omega]$ and the two translation homotopies converge in the stated ambient, then

$$Z_{\text{ch}}^{\text{der}}(A)$$

is promoted from the closed colour of $SC_{\text{ch}, \text{top}}$ to an E_3^{top} -factorisation algebra. For class M , the raw direct-sum statement is not a theorem; the manuscript itself says class M lives in the weight-completed pro-ambient unless finite propagation is separately proved. main

II. The chiral derived centre must be defined in the chiral bimodule category

The ordinary associative formula

$$RHom_{A \otimes A^{op}}(A, A)$$

is not the definition of the chiral centre. It is only a model after one has chosen a chiral enveloping algebra or a Ran-space bimodule presentation.

The definition must be:

$$Z_{\text{ch}}^{\text{der}}(A) := \mathbb{R}\text{End}_{A\text{-Bimod}^{\text{ch}}(X)}(A).$$

Here $A\text{-Bimod}^{\text{ch}}(X)$ is the derived category of chiral A -bimodules on $\text{Ran}(X)$. Equivalently, an A -bimodule is a right $D_{\text{Ran}(X)}$ -module M with compatible left and right chiral operations along the partial diagonals

$$j_* j^*(A \boxtimes M) \rightarrow \Delta_* M, \quad j_* j^*(M \boxtimes A) \rightarrow \Delta_* M.$$

On the chirally Koszul locus, the chiral bar construction gives a free chiral A -bimodule resolution

$$\Omega_{\text{ch}} B^{\text{ch}}(A) \sim A.$$

Therefore

$$Z_{\text{ch}}^{\text{der}}(A) \simeq \text{Hom}_{A\text{-}A}^{\text{ch}}(\Omega_{\text{ch}} B^{\text{ch}}(A), A) \simeq \text{Hom}(B^{\text{ch}}(A), A) = C_{\text{ch}}^{\bullet}(A, A).$$

This is already latent in the manuscript, where the bar resolution is used to identify the chiral derived centre with chiral Hochschild cochains and where the brace operations realise the $SC_{\text{ch}, \text{top}}$ -action.

main

The brace operations must be displayed, not merely named. For

$$f \in C_{\text{ch}}^p(A, A), \quad g_i \in C_{\text{ch}}^{q_i}(A, A),$$

define

$$f\{g_1, \dots, g_r\} = \sum_{\Gamma \in \text{Ins}(p; q_1, \dots, q_r)} \pm \mu_{\Gamma}(f; g_1, \dots, g_r),$$

where Γ ranges over ordered insertions of the g_i into the chiral inputs of f , and

$$\mu_{\Gamma} = \int_{\text{Conf}_{\Gamma}(X) \times E_1(r)} \omega_{\Gamma} \text{ OPE}_{\Gamma}.$$

The differential satisfies

$$d(f\{g\}) - (df)\{g\} - (-1)^{|f|} f\{dg\} = f \circ g - (-1)^{(|f|-1)(|g|-1)} g \circ f,$$

and the induced bracket

$$[f, g]_G = f \circ g - (-1)^{(|f|-1)(|g|-1)} g \circ f$$

is the chiral Gerstenhaber bracket.

The bulk-boundary action is the map

$$C_{\text{ch}}^{\bullet}(A, A) \otimes A \rightarrow A$$

given by insertion of a Hochschild cochain into a boundary field. Thus

$$(Z_{\text{ch}}^{\text{der}}(A), A)$$

is the object carrying the Swiss-cheese structure. The bar coalgebra is a resolution; it is not the bulk. The manuscript states this distinction in its own preface: cobar, Verdier duality, and Hochschild cochains are three distinct projections of the same bar coalgebra, and identifying them collapses the architecture.

main

III. $SC_{\text{ch},\text{top}}$ is a directed two-coloured dioperad

The manuscript cannot call $SC_{\text{ch},\text{top}}$ a standard operad on one page and a dioperad on another. The mathematical object is:

$$\begin{aligned}\text{Col} &= \{\text{ch}, \text{top}\}, \\ SC_{\text{ch},\text{top}}(\text{ch}^{\times k}; \text{ch}) &= FM_k(\mathbb{C}), \\ SC_{\text{ch},\text{top}}(\text{ch}^{\times k}, \text{top}^{\times m}; \text{top}) &= FM_k(\mathbb{C}) \times E_1(m), \\ SC_{\text{ch},\text{top}}(\dots, \text{top}, \dots; \text{ch}) &= \emptyset.\end{aligned}$$

Composition is Fulton–MacPherson insertion in the chiral colour and interval insertion in the topological colour. Symmetric equivariance exists within each colour. No equivariance exchanges ch and top . The manuscript itself states that this is a strict two-coloured topological dioperad in Gan's sense, that inter-colour equivariance is absent, and that Dunn additivity does not apply before topologisation. main

Therefore every occurrence of

“the operad $SC_{\text{ch},\text{top}}$ ”

must become

“the directed two-coloured topological dioperad $SC_{\text{ch},\text{top}}$ ”

unless a Boardman–Vogt operadic envelope has been explicitly constructed.

The topological E_3 statement is not

$$SC_{\text{ch},\text{top}} \simeq E_3.$$

The correct implication is:

$$SC_{\text{ch},\text{top}}\text{-Alg} \quad + \quad \partial_z = [Q, h_z], \quad \partial_{\bar{z}} = [Q, h_{\bar{z}}] \quad \implies \quad E_3^{\text{top}}\text{-Alg}$$

on the closed colour.

In physical language, the chiral collision plane becomes topological only after the stress tensor is Q -exact. The real line was already topological. Thus

$$E_2^{\text{top}} \times E_1^{\text{top}} \simeq E_3^{\text{top}}$$

appears after topologisation, not before.

IV. The master theorem must be split into algebraic, BV, and topological theorems

Theorem A: algebraic centre theorem

Let A be a chirally Koszul chiral algebra on a smooth curve X . Then

$$Z_{\text{ch}}^{\text{der}}(A) = \mathbb{R}\text{End}_{A\text{-Bimod}^{\text{ch}}(X)}(A)$$

is computed by the chiral Hochschild complex

$$C_{\text{ch}}^{\bullet}(A, A) = \text{Hom}(B^{\text{ch}}(A), A).$$

The pair

$$(C_{\text{ch}}^{\bullet}(A, A), A)$$

is an algebra over $W(SC_{\text{ch},\text{top}})$. The closed colour is the chiral brace algebra on $C_{\text{ch}}^{\bullet}(A, A)$. The open colour is the E_1 -ordered boundary algebra. The mixed colour is the evaluation action of chiral Hochschild cochains on boundary fields.

Theorem B: BV realisation theorem

Let

$$(A, \omega, BL, \Xi)$$

be an object of $HTReal_X^{Kosz, \omega, BL}$. Then perturbative Costello–Gwilliam quantisation produces a factorisation algebra

$$\text{Obs}_{\Xi}^q$$

on $X \times \mathbb{R}_{\geq 0}$ with boundary and bulk restrictions

$$\text{Obs}_{\partial, \Xi}^q \simeq A, \quad \text{Obs}_{\text{bulk}, \Xi}^q \simeq Z_{\text{ch}}^{\text{der}}(A).$$

The first equivalence is BL . The second equivalence is χ_{bulk} . The two maps must be part of Ξ , or proved by a BV calculation in the example.

Theorem C: topologisation theorem

If the conformal vector satisfies

$$T_{\omega} = [Q, G_{\omega}]$$

and the translation operators in the holomorphic directions satisfy

$$\partial_z = [Q, h_z], \quad \partial_{\bar{z}} = [Q, h_{\bar{z}}],$$

then $\text{Obs}_{\text{bulk}, \Xi}^q$ is locally constant on $X \times \mathbb{R}$. Hence

$$\text{Obs}_{\text{bulk}, \Xi}^q$$

is an E_3 -algebra, and

$$Z_{\text{ch}}^{\text{der}}(A)$$

inherits an E_3^{top} -factorisation structure through χ_{bulk} .

For class $\bar{M}$, all maps must be interpreted in the completed ambient where the MC series converges. The manuscript itself states that the raw direct-sum theorem is not available in class $\bar{M}$. main

**V. The proof must contain the actual BV construction**

Costello will not accept “there exists a QFT” without fields, local functionals, gauge fixing, and the QME.

For each standard family, the manuscript must write the following data.

For Chern–Simons type theories on a three-manifold M ,

$$\mathcal{E}_{CS} = \Omega^{\bullet}(M, \mathfrak{g})[1], \quad Q = d,$$

$$\omega_{BV}(\alpha, \beta) = \int_M \langle \alpha, \beta \rangle,$$

$$S_{CS}(A) = \int_M \left(\frac{1}{2} \langle A \wedge dA \rangle + \frac{1}{6} \langle A \wedge [A \wedge A] \rangle \right).$$

For holomorphic Chern–Simons on a Calabi–Yau threefold Y ,

$$\mathcal{E}_{hCS} = \Omega^{0, \bullet}(Y, \mathfrak{g})[1], \quad Q = \bar{\partial},$$

$$S_{hCS}(\mathcal{A}) = \int_Y \Omega_Y \wedge \left(\frac{1}{2} \langle \mathcal{A} \wedge \bar{\partial} \mathcal{A} \rangle + \frac{1}{6} \langle \mathcal{A} \wedge [\mathcal{A} \wedge \mathcal{A}] \rangle \right).$$

The manuscript’s Appendix B correctly says these two theories are not equal objects: CFG 2026 is ordinary topological Chern–Simons, while the Vol II lane is Dolbeault BV/hCS/chiral avatar, and the two agree only after named comparison functors and hypotheses. main

Therefore Chapter 34 and Appendix B must never assert equality of the CFG factorisation algebra with the Vol II hCS factorisation algebra. The correct comparison diagram is:

$$\mathrm{Obs}_{CS}^q(M) \xrightarrow{\mathrm{CG\,BV}} \mathrm{FactAlg}_{E_3}(M) \xrightarrow{\mathrm{recognition}} \mathrm{LocConstFactAlg}(M)$$

and separately

$$\mathrm{Obs}_{hCS}^q(Y) \xrightarrow{\mathrm{Dolbeault\,BV}} \mathrm{FactAlg}_{\mathrm{hol}}(Y) \xrightarrow{SpCh_{\Sigma,C}} \mathrm{ChirAlg}(C) \xrightarrow{Z_{\mathrm{ch}}^{\mathrm{der}}} SC_{\mathrm{ch,top}}\text{-Alg}.$$

No arrow in this diagram is an identity unless the corresponding comparison theorem has been proved.

VI. The A_∞ calculus must be written as Stokes calculus

Kontsevich will not accept “the A_∞ -relations follow from FM boundaries” without the forms, residues, and signs.

For a chiral A_∞ -algebra, the operations are

$$m_n : A^{\otimes n} \longrightarrow A((z_1 - z_n)) \cdots ((z_{n-1} - z_n))[2 - n].$$

Let

$$\omega_{ij} = d \log(z_i - z_j).$$

The Arnold relation on $\mathrm{Conf}_3(\mathbb{C})$,

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0,$$

is the codimension-two Stokes identity giving the quadratic A_∞ -relation and the pre-Lie identity for braces. The manuscript already states this relation in the brace appendix; that calculation must be promoted into the main proof, because it is the first place where the whole machine becomes mathematical rather than verbal. main

The relation

$$D_A^2 = 0$$

must be written as

$$\sum_{\substack{i+j=n+1 \\ 0 \leq s \leq n-j}} (-1)^{\epsilon(s,j)} m_i(a_1, \dots, a_s, m_j(a_{s+1}, \dots, a_{s+j}), a_{s+j+1}, \dots, a_n) = 0,$$

with

$$\epsilon(s, j) = j \sum_{r=1}^s |a_r| + s(j-1)$$

or whatever sign convention the manuscript fixes in Appendix A. The sign convention must be used consistently in the FM orientation calculation.

The sentence “nonconsecutive divisors vanish by ordering” must only appear in the open planar E_1 projection. In the closed chiral colour $FM_n(\mathbb{C})$, every labelled collision divisor exists.

VII. Etingof-level quantum-group content must be separated from normalization choices

For affine Kac–Moody,

$$A = V_k(\mathfrak{g}), \quad k \neq -h^\vee.$$

The KZ connection on n -point conformal blocks is

$$\nabla_i = \partial_{z_i} - \frac{1}{k + h^\vee} \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}.$$

The infinitesimal braiding is

$$r_{\text{KZ}}(z) = \frac{\Omega}{(k + h^\vee)z}.$$

If the manuscript also uses the trace-form convention

$$r_{\text{tr}}(z) = \frac{k \, \Omega_{\text{tr}}}{z},$$

then it must state the linear change of invariant form relating Ω_{tr} and Ω . The two formulas cannot appear as if they were the same r -matrix.

The line category statement must be:

$$\mathcal{C}_{\text{line}}^{\text{eval}}(V_k(\mathfrak{g})) \simeq \text{Rep}_q^{\text{eval}}(\mathfrak{g}), \quad q = \exp\left(\frac{\pi i}{k + h^\vee}\right),$$

under the Kazhdan–Lusztig/Drinfeld–Kohno comparison hypotheses. The manuscript already states the q -formula for affine Kac–Moody line categories, but the proof must go through the monodromy of KZ, not through an unnamed “spectral braiding” slogan. main

The Yang–Baxter equation must be derived from the cubic MC equation:

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0.$$

The manuscript has exactly this displayed mechanism in the derived-centre/bar-resolution passage; it should be part of the main theorem proof rather than buried downstream. main

VIII. Gaiotto-level boundary/defect physics requires a slab theorem

The bulk-boundary-line triangle must be a theorem about a half-space or slab:

$$X \times \mathbb{R}_{\geq 0} \quad \text{or} \quad X \times [0, 1].$$

Let

$$\mathcal{B} = \text{Obs}_{\partial_0}, \quad \mathcal{B}^\dagger = \text{Obs}_{\partial_1}, \quad \mathcal{Z} = \text{Obs}_{\text{bulk}}, \quad \mathcal{C}_{\text{line}} = \text{Line}(X \times \mathbb{R}_{>0}).$$

The theorem must state:

$$\begin{aligned} \mathcal{Z} &\simeq Z_{\text{ch}}^{\text{der}}(\mathcal{B}), \\ \mathcal{C}_{\text{line}} &\simeq A_{\text{line}}^! \text{-mod} \end{aligned}$$

on the chirally Koszul locus, and

$$R_{L_1, L_2}(z) : L_1(z_1) \otimes L_2(z_2) \rightarrow L_2(z_2) \otimes L_1(z_1)$$

is the bulk-mediated braiding obtained by transporting two parallel topological lines through a chiral collision in the X -direction.

The manuscript already has the exact slogan that the level C chart algebra is the boundary, the level Z derived chiral centre is the bulk, and the level A line algebra is the line-operator content. That sentence should become a theorem with maps, not remain an architectural reading. main

IX. Polyakov/Witten-level gravity requires the chiral/full split

The statement

$$A = \text{Vir}_c \quad \Rightarrow \quad \text{3d quantum gravity}$$

is false unless it is read as a chiral boundary sector.

The correct physical theorem is:

Let M be an oriented three-manifold with asymptotic boundary X . Euclidean or Lorentzian AdS_3 gravity with Brown–Henneaux boundary conditions has Chern–Simons variables

$$A = \omega + \frac{1}{\ell} e, \quad \bar{A} = \omega - \frac{1}{\ell} e,$$

and action

$$S_{\text{grav}} = S_{CS}[A] - S_{CS}[\bar{A}].$$

The Chern–Simons level is

$$k_{CS} = \frac{\ell}{4G},$$

and the boundary central charge is

$$c = 6k_{CS} = \frac{3\ell}{2G}.$$

Thus a single Virasoro algebra Vir_c is one chiral half of the asymptotic symmetry algebra. The full non-chiral boundary algebra is

$$\text{Vir}_c \boxtimes \text{Vir}_c.$$

The physical partition function requires a choice of real form, boundary conditions, integration cycle, modular invariant Hilbert space, and saddle prescription:

$$Z_{\text{grav}}(\tau, \bar{\tau}) = \sum_{\gamma \in \Gamma_\infty \backslash SL_2(\mathbb{Z})} Z_{\text{saddle}}(\gamma \cdot \tau, \gamma \cdot \bar{\tau})$$

only after the Maloney–Witten saddle problem or its replacement has been supplied.

The manuscript's own Chapter 41 language is already safer: the Virasoro HT model is the value of the functor at a Brown–Henneaux Virasoro datum, while class M is weight-completed and the strict E_3 lift requires the stated hypotheses. main

Therefore the manuscript should say:

$$Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$$

is the chiral holomorphic-topological bulk algebra associated to the Brown–Henneaux Virasoro datum. It is not, by itself, the full gravitational path integral.

X. The abstract should be replaced by this

Let X be a smooth complex curve and let A be a chirally Koszul chiral algebra on X . The chiral derived centre is

$$Z_{\text{ch}}^{\text{der}}(A) := \mathbb{R}\text{End}_{A\text{-Bimod}^{\text{ch}}(X)}(A).$$

On the Koszul locus the chiral bar resolution gives

$$Z_{\text{ch}}^{\text{der}}(A) \simeq C_{\text{ch}}^\bullet(A, A) = \text{Hom}(B^{\text{ch}}(A), A).$$

The pair

$$(Z_{\text{ch}}^{\text{der}}(A), A)$$

is naturally an algebra over the directed two-coloured chiral-topological Swiss-cheese dioperad $SC_{\text{ch}, \text{top}}$, whose operation spaces are

$$SC_{\text{ch}, \text{top}}(\text{ch}^{\times k}; \text{ch}) = FM_k(\mathbb{C}),$$

$$SC_{\text{ch},\text{top}}(\text{ch}^{\times k}, \text{top}^{\times m}; \text{top}) = FM_k(\mathbb{C}) \times E_1(m),$$

$$SC_{\text{ch},\text{top}}(\dots, \text{top}, \dots; \text{ch}) = \emptyset.$$

The closed colour is the chiral brace algebra on $C_{\text{ch}}^\bullet(A, A)$; the open colour is the E_1 -ordered boundary algebra; the mixed colour is the evaluation action of chiral Hochschild cochains on A .

A physical holomorphic-topological theory is obtained only after adding an HT realisation datum

$$\Xi = (\mathcal{E}, Q, I, \omega_{BV}, \mathcal{L}_\partial, P_L, \eta_\partial, \chi_{\text{bulk}}, G_\omega, h_{\text{HT}})$$

solving the renormalised quantum master equation and comparing the boundary and bulk observables with A and $Z_{\text{ch}}^{\text{der}}(A)$. The resulting functor is

$$\Phi_{\text{hol}} : HTReal_X^{\text{Kosz}, \omega, BL} \longrightarrow HTQFT_{X \times \mathbb{R}_{\geq 0}},$$

and satisfies

$$\text{Obs}_\partial^q(\Phi_{\text{hol}}(A, \Xi)) \simeq A, \quad \text{Obs}_{\text{bulk}}^q(\Phi_{\text{hol}}(A, \Xi)) \simeq Z_{\text{ch}}^{\text{der}}(A).$$

If the conformal vector is Q -exact and the holomorphic translations are Q -exact in the chosen chain model, the bulk becomes an E_3^{top} -factorisation algebra. For class $\bar{M}$ this statement is made in the weight-completed pro-ambient unless finite propagation on the original direct-sum complex is proved.

For $A = V_k(\mathfrak{g})$, the line category on the evaluation sector recovers the Drinfeld–Kohno/Kazhdan–Lusztig quantum group with

$$q = \exp\left(\frac{\pi i}{k + h^\vee}\right).$$

For $A = \text{Vir}_c$, the construction gives the chiral Brown–Henneaux holomorphic-topological sector of three-dimensional gravity; the full non-chiral gravitational path integral requires $\text{Vir}_c \boxtimes \text{Vir}_c$, a real form, modular invariance, and a saddle prescription.

XI. The high-level claims need these exact corrections

Replace

“non-logarithmic C_2 -cofinite standard landscape containing generic Virasoro and generic W_3 ”

by

“the standard non-critical tempered landscape, with C_2 -cofinite rational/lisse loci separated from generic Virasoro

The manuscript itself later separates generic Virasoro and principal $\bar{W}_N$ tempering from logarithmic triplet questions and irrational-coset questions, so the abstract’s C_2 -cofinite phrasing is too coarse.

main

Replace

“ $SC_{\text{ch},\text{top}}$ is the operadic record”

by

“ $SC_{\text{ch},\text{top}}$ is the directed two-coloured dioperadic record.”

Replace

“the bar is an $SC_{\text{ch},\text{top}}$ -coalgebra”

by

$B^{\text{ch}}(A)$ is an E_1 dg coassociative coalgebra resolving A ,

and

$(Z_{\text{ch}}^{\text{der}}(A), A)$ is the $SC_{\text{ch},\text{top}}$ -algebra.

The manuscript explicitly says this distinction: the bar complex is the E_1 chiral coassociative coalgebra, while the Swiss-cheese structure lives on the chiral derived-centre pair. main

Replace

“CFG2026 matches Vol II”

by

“CFG2026 supplies the ordinary topological Chern–Simons factorisation lane; Vol II supplies the Dolbeault hCS/

The appendix already says this precisely. main

Replace

“3d quantum gravity is $A = \text{Vir}_c$ ”

by

“ $A = \text{Vir}_c$ is the chiral Brown–Henneaux boundary datum for the HT avatar of 3d gravity.”

XII. The manuscript’s ideal form

The ideal manuscript has this theorem chain:

BD chiral algebra $\Rightarrow$ chiral bar resolution $\Rightarrow$ chiral Hochschild centre $\Rightarrow SC_{\text{ch,top}}$ -bulk-boundary algebra

$$SC_{\text{ch,top}}\text{-bulk-boundary algebra} + \Xi_{\text{BV}} \Rightarrow \text{HT factorisation algebra on } X \times \mathbb{R}_{\geq 0}$$

$$T = [Q, G] + \partial_z = [Q, h_z] + \partial_{\bar{z}} = [Q, h_{\bar{z}}] \Rightarrow E_3^{\text{top}}\text{-bulk}$$

$$A = V_k(\mathfrak{g}) \Rightarrow \text{KZ monodromy} \Rightarrow \text{Rep}_q(\mathfrak{g})$$

$$A = \text{Vir}_c \Rightarrow \text{chiral Brown–Henneaux HT sector} \subset \text{full non-chiral AdS}_3 \text{ gravity only after the missing gravitation}$$

That is the manuscript that can keep the ambition. The existing PDF already contains many of the right components, but they are not yet arranged as typed mathematics and typed physics. The healing is to make the abstract, theorem statements, and proof bodies obey the same mathematical distinctions that the later sections already know: $SC_{\text{ch,top}}$ is a dioperad, the bar is not the bulk, $Z_{\text{ch}}^{\text{der}}$ is a Ran/chiral bimodule endomorphism object, Φ_{hol} requires Ξ , E_3 requires Q -exact topologisation, CFG is a comparison lane rather than an equality, and Virasoro gives the chiral gravity sector rather than the full gravitational path integral.

        Sources

Is this conversation helpful so far?   